```
  (https://remotexy.com/en/library/)
 * Software Platform: Arduino IDE 1.8.13
  (Win10x64) & Android 10
 */

#define REMOTEXY_MODE__SOFTSERIAL
#include <SoftwareSerial.h>
#include <RemoteXY.h>

#define REMOTEXY_SERIAL_RX 2
#define REMOTEXY_SERIAL_TX 3
#define REMOTEXY_SERIAL_SPEED 9600

#pragma pack(push, 1)
uint8_t RemoteXY_CONF[] =
  { 255,1,0,0,0,11,0,10,130,0,
  4,128,13,20,80,22,108,16 };


struct {

  int8_t slider_1;

  uint8_t connect_flag;

} RemoteXY;
#pragma pack(pop)

#define PIN_SLIDER_1 9

void setup()
{
  RemoteXY_Init ();
  pinMode (PIN_SLIDER_1, OUTPUT);

}

void loop()
{
  RemoteXY_Handler ();
  analogWrite(PIN_SLIDER_1, RemoteXY.
  slider_1*2.55);

}
```

Once the hardware setup is complete, the Android application RemoteXY can be downloaded and installed in the Android OS, which will manage the connection between the Arduino and the smartphone. RemoteXY is a bit different from similar Android applications for Arduino you may be familiar with.

You do not have to connect to the Internet to link your smartphone and Arduino setup with Bluetooth. The software has all the needed instructions and the user interface (slider). The Arduino setup smoothly sends the interface and configurations data to the RemoteXY app in your smartphone and then the smartphone app interprets the interface offline without the need to connect to the Internet!

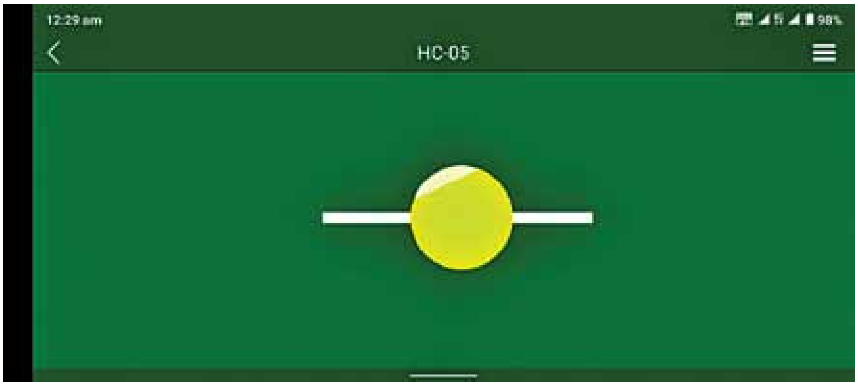

Fig. 4: Slider of RemoteXY app

RemoteXY Android app can be downloaded from *https://play.google.com/store/apps/details?id=com.shevauto.remotexy.free*

Follow the steps below for proper operation of the lamp:

1. Power the Arduino setup using a 12V DC adaptor.
2. Start and run the smartphone RemoteXY application.
3. Turn on Bluetooth connectivity.
4. Press the new connection '+' button in the top panel of the application as shown in Fig. 3.
5. In the window that opens, select 'Bluetooth' connection.
6. Select your HC-05 device.
7. In the new window, enter password 1234 (or 0000) to establish the connection.
8. Start controlling/regulating the brightness of the LED light source with the slider!

Fig. 4 shows the screenshot taken by the author during testing of the prototype with an Android smartphone (Nokia 5.3/Android 10).

If there is a connection problem, you might get error message: "The connection is established, but the device does not respond: Time out error: no response!" If you receive this message, recheck your Arduino setup (hardware and software) and/or the smartphone's Bluetooth connectivity.

The author built his quick prototype using an Arduino prototype shield (for his convenience). The white LED module is in fact a pair of 12V DC cool daylight-white LED plates (x3 lamps per plate) wired in parallel. Its total current consumption, as observed by the author, is 200mA at 12V DC.

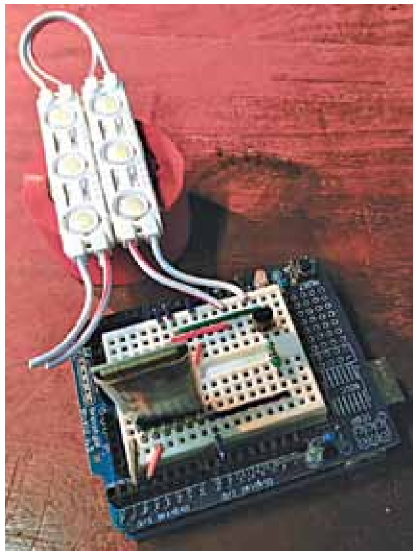
Fig. 5: Bluetooth module with 12V white LED

To obtain HC-05 compatible Bluetooth module you may check *https://www.rhydolabz.com/wireless-bluetooth-ble-c-130_132/hc05-compatible-bluetooth-module-with-base-board-p-2842.html* and for the 12V white LED module check *https://www.amazon.in/PickTheDeal-Waterproof-Injection-Module-Warm-White/dp/B07MX4DNDQ/*

The author's project works quite well but has not passed the prototype stage yet. Any additional ideas will be appreciated! EFY

EFY Note
The source code of this project is available for free download at *source.efymag.com*


T.K. Hareendran is an electronics designer, hardware beta tester, technical author, and product reviewer